

Mallarpu Aishwarya

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PROFESSIONAL SUMMARY

Detail-oriented Computer Science undergraduate specializing in Data Science with strong foundations in Data Structures, Algorithms, DBMS, Operating Systems, and Object-Oriented Programming. Demonstrated ability to design scalable architectures, optimize system performance, and implement structured problem-solving methodologies.

EXPERIENCE

Indian Institute of Technology Kharagpur – Department of CSE

May 2025 - Jun 2025

Summer Intern – Database Management & Web Service Development

- Designed and optimized relational database schemas, reducing query latency by approximately 20% through indexing and normalization techniques.
- Developed RESTful APIs serving structured JSON responses, handling 500+ requests per day during internal testing.
- Improved database query performance by refactoring inefficient joins and reducing computational complexity from $O(n^2)$ to $O(n \log n)$ for key data retrieval operations.

PROJECTS

AI Systems Engineering – NLP & Retrieval-Augmented Generation (RAG)

- Built end-to-end NLP applications using HuggingFace Transformers for text generation, Named Entity Recognition (NER), sentiment analysis, and zero-shot classification across multilingual datasets.
- Developed a Gradio-based interactive Text Analyzer supporting real-time local model inference with modular UI components using `gr.Blocks`.
- Designed and evaluated a Retrieval-Augmented Generation (RAG) pipeline processing 1,000+ documents, improving retrieval relevance by 18% using Mean Reciprocal Rank (MRR) and keyword coverage metrics.
- Implemented query rewriting and LLM-as-Judge evaluation strategies, increasing response accuracy and contextual grounding by approximately 15%.

Backend Web Service Development

- Developed RESTful APIs using Flask/FastAPI enabling CRUD operations for structured data processing across 10+ modular endpoints.
- Integrated MySQL and MongoDB for persistent storage, ensuring data validation and consistency across 5,000+ records. Applied Git-based version control and followed SDLC best practices, improving development efficiency and reducing integration errors during iterative builds.

EDUCATION

C. V. Raman Global University, Bhubaneswar

2023 - 2027

B.Tech in Computer Science & Engineering (Data Science)

CGPA: 7.65

Relevant Coursework: Data Structures, Algorithms, DBMS, Operating Systems, Computer Networks, Machine Learning

TECHNICAL SKILLS

Programming: Python, C

Core CS: Data Structures, DBMS, Computer Networks

Frontend: HTML5, CSS, JavaScript

AI / Data Science: NumPy, Pandas, HuggingFace Transformers, NLP, RAG Concepts

Tools & Practices: GitHub, Postman, CI/CD

CERTIFICATIONS

- JPMorgan Chase – Software Engineering Virtual Experience (Forage)
- Deloitte Australia – Data Analytics Virtual Experience (Forage)
- IBM – Introduction to Cyber Security Tools & Cyber Attacks
- Palo Alto Networks – Cybersecurity Foundations
- Infosys Springboard – Deep Learning, OpenAI GPT Models, Agile Scrum

ACHIEVEMENTS

- Cleared Smart India Hackathon – Internal Round (2025)
- 2nd Prize – CTF Competition, IIIT Bhubaneswar (2025)
- Finalist – CID Crime Branch Odisha Police × IIT Bhubaneswar Hackathon (2025)